FSVLM: A Vision-Language Model for Remote Sensing Farmland Segmentation

Haiyang Wu, Zhuofei Du, Dandan Zhong, Yuze Wang, and Chao Tao, *Member, IEEE*

***Abstract*— Existing visual deep learning paradigms, which are based on labels, struggle to capture the intricate interrelationships between farmland and its surrounding environment and fail to account for temporal variations associated with the phenological cycle. These limitations lead to omissions and confusion in the recognition process, greatly impacting the accuracy and efficiency of farmland recognition. Language can accurately depict the spatial attributes of farmland, profoundly express the unique phenological landscapes of farmland that change with seasons and growth stages, and express the intricate interactions between these changes and environmental factors. This capability can address the deficiencies of label-based visual deep learning in understanding the complex features of farmland. This study explored, for the first time, the application of language-guided vision-language models (VLMs) for farmland segmentation. First, as current VLM research lacks a dedicated farmland image text (FIT)pair dataset, this study constructed an FIT dataset in two steps. Step 1, designed a semi-automatic text description annotation framework for farmland images based on 12 key factors influencing farmland segmentation. Step 2, used the framework to construct the FIT dataset. Then, a VLM for farmland segmentation (FSVLM) was designed by combining a semantic segmentation model with a multimodal large language model (LLM). Comparative experiments demonstrated that the proposed method outperforms existing farmland segmentation methods in both generalization and segmentation accuracy. In addition, a series of ablation experiments were conducted to examine the impacts of language descriptions of different semantic levels on the model's farmland information extraction performance.**

***Index Terms*— Deep learning, farmland segmentation, image-text dataset, semantic segmentation, vision-language model (VLM).**

I. Introduction

AGRICULTURAL development constitutes a fundamental pillar of economic advancement, intricately linked to the stability and prosperity of human societies. The efficient and precise delineation of arable land resources has become essential for informed decision-making and the advancement of agricultural modernization [1]. Remote sensing technology has demonstrated great potential and advantages in monitoring and managing arable land resources over a wide area; therefore, in-depth research and application of remote sensing technology-based methods for farmland resource extraction are urgent and necessary.

In recent years, the advent of deep learning technology, especially the successful application of convolutional neural networks (CNNs) in remote sensing image processing [2], [3], [4], has brought new opportunities for accurate extraction of farmland information from remote sensing images [5]. CNNs can autonomously learn and extract multilevel, multiscale features from remote sensing images, including spectral information, spatial context information, and advanced semantic information. This can reduce the "salt and pepper" effect, thereby improving the accuracy and robustness of classification results. Consequently, an increasing number of studies have used deep learning-based methods for remote sensing farmland segmentation [6], [7], [8], [9], [10]. For example, a method for detecting farmland boundary was proposed, which integrates a fully CNN with residual structures [8]. This method was proved to be effective in identifying the boundaries of farmland. The incorporation of attention mechanisms into CNNs has, moreover, led to the development of a dual multiscale attention neural network [10], which has been used for farmland segmentation in southern Xinjiang, China, yielding enhanced segmentation accuracy. Additionally, a multitask-based deformable neural network method for farmland boundary segmentation was proposed [11], and it achieved better segmentation performance on GF-2 imagery than the DeepLab v3 model [12].

Because of the significant imbalance in the distribution of different ground objects within the sample dataset, directly using neural networks to predict farmland boundary pixels in remote sensing images often yields poor results. In order to address this problem, Kuang et al. [13] explored a joint approach of multisource remote sensing data for extracting farmland boundaries and demonstrated the feasibility of combining SAR remote sensing imagery to optimize farmland boundary extraction. In addition, existing semantic segmentation-based methods find it difficult to extract fine and complete farmland edges. In order to solve this problem, a farmland edge postprocessing method was designed, effectively extracting closed farmland boundary lines from Google Earth images [14]. Besides, a multiscale edge network was designed to mitigate edge blurring in farmland segmentation tasks [15].

Farmland exhibits notable spatiotemporal attributes, and the farmland environments are complex, posing challenges

Received 8 September 2024; revised 31 December 2024; accepted 20 January 2025. Date of publication 23 January 2025; date of current version 4 February 2025. This work was supported in part by the Natural Science Foundation of Hunan for Distinguished Young Scholars under Grant 2022JJ10072 and in part by the National Natural Science Foundation of China under Grant 42471419 and Grant 42171376. *(Corresponding author: Chao Tao.)*

The authors are with the School of Geosciences and Info-Physics, Central South University, Changsha 410083, China (e-mail: kingtaochao@csu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3532960

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:58:22 UTC from IEEE Xplore. Restrictions apply.

for current label learning-based deep learning paradigms in representing these spatiotemporal features. This impedes efficient farmland segmentation. Specifically, the distribution and morphology of farmland vary with seasonal changes. For example, during the sowing and harvesting periods, the appearance of farmland differs significantly. Farmland is also influenced by local climate, artificial structures, water resources, and vegetation distribution. All these factors contribute to the spatiotemporal features of farmland. Label-based deep learning paradigms can learn the shape and texture features of farmland and surrounding environment but fail to represent the internal logic and correlation between them [16]; furthermore, they are inadequate in capturing the phenological features of farmland, which are influenced by seasonal changes, climate fluctuation and the evolution of crop growth cycle. For example, farmland with crops exhibits significant visual differences compared to fallow land. Models that solely rely on label-based learning tend to misclassify fallow land as nonfarmland because they lack a deep understanding or consideration of key information, such as the phenological changes and the crop growth cycle. These limitations lead to inaccuracies and limitations in the segmentation results.

Language, expressing rich semantic and logical relationships, endow models with the ability to understand the inherent relationships between farmland and surrounding environment [17]. By using language descriptive texts to guide vision models for farmland segmentation, the model's capabilities can be elevated from mere "visual recognition" to a more nuanced "visual understanding" [18]. This approach effectively addresses and mitigates the limitations of existing methods in farmland segmentation. Specifically, language can provide rich prior knowledge by describing the season, shape, and distribution of farmland, as well as environmental features such as surrounding artificial structures, water resource status, and vegetation layout. With this previous knowledge, the model can better describe the internal relationships between farmland and its surrounding environment, mitigating the segmentation challenges posed by the temporal and spatial variations and achieving more efficient and accurate farmland segmentation performance.

With the ongoing advancements in model architecture and large language models (LLMs) [19], combining language modalities with image modalities to design highly robust vision-language models (VLMs) with significant inference capabilities has become a research hotspot [20]; however, current research on applying VLMs to farmland segmentation is nascent, and there is an urgent need to explore the role of different descriptive texts within the model. Additionally, existing VLMs lack farmland image-text (FIT) pair datasets. In order to address these issues, this work first identifies the factors that pose challenges for farmland segmentation through a comprehensive analysis of existing research and extensive observation of a large number of farmland remote sensing images. Subsequently, a semi-automatic annotation framework was designed to construct a FIT pair dataset. Then, a VLM for farmland segmentation (FSVLM) was designed by combining the segmentation model with a multimodal LLM. Finally, to identify the most effective descriptive texts for farmland segmentation in a VLM, this study compares the performance of FSVLM with different language descriptions in farmland segmentation.

The contributions of this article are as follows:

1) This study identified 12 key elements that describe the attributes of farmland and its surrounding environment. Based on these elements, a semi-automatic text description annotation framework was designed; furthermore, the framework was used to construct a FIT pair dataset.
2) The FSVLM was designed by combining a segmentation model with a multimodal LLM. This study is the first to explore the application of FSVLMs. Experiments demonstrated that the proposed method outperforms label-based learning approaches in generalization and segmentation accuracy.
3) This study thoroughly analyzed the impacts of different types of descriptive texts on the effectiveness of the FSVLM in farmland segmentation. We found that descriptive texts focusing on farmland have higher model accuracy and efficiency in farmland segmentation tasks than those focusing on the surrounding environment. This discovery provides reference for the construction of subsequent FIT pair datasets.

The rest of this article is arranged as follows. Section II reviews the related work. Section III thoroughly discusses the proposed method from two aspects: the construction of the FIT dataset and the principle and implementation of FSVLM. In Section IV, comparative experiments and model generalization tests are done, demonstrating that the proposed method has great segmentation performance and generalization in farmland identification. Additionally, ablation experiments are performed to examine the impacts of different types of descriptive texts on the effectiveness of FSVLM in farmland segmentation. The conclusion is presented in Section V.

II. Related Work

A. Remote Sensing Farmland Segmentation

Recent advancements in deep learning technology have led to significant improvements in semantic segmentation models and their application in farmland segmentation. Conventional segmentation models often struggle with accurately delineating farmland boundaries. For addressing this problem, fully convolutional networks combined with grouping algorithms were employed [21]. This strategy has significantly enhanced the precision of farmland segmentation, providing robust support for precision agriculture management. Graph attention mechanisms were, furthermore, integrated with CNN, and the resulted method was successfully applied to hyperspectral image analysis, enabling fine-grained differentiation of farmland planted with different crop types [22]. This approach not only enhances the model's understanding of complex farmland environments but also facilitates precise monitoring of crop planting structures. An ensemble learning strategy that averages the prediction results of four deep semantic

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:58:22 UTC from IEEE Xplore. Restrictions apply.

segmentation models was proposed by Cheng et al. [23] to get even better segmentation performance. This strategy effectively mitigates the classification biases that may arise from using a single model. Experiments demonstrated the superiority of this ensemble model in farmland extraction tasks, underscoring its potential to enhance the stability and reliability of segmentation outcomes. Additionally, [24] addressed the challenge of large-scale farmland extraction by adopting an enhanced version of the modified pyramid scene parsing network combined with a U-Net model, successfully achieving precise farmland extraction.

In spite of the significant progress achieved by the above methods, they are all constrained by the supervised learning paradigm based on labels. This paradigm heavily relies on precise and complete labeled data yet ignores the deep underlying logic and interrelationships between farmland and its surrounding environment. As a result, these methods struggle to fully capture the dynamic changes and the reciprocal influences between farmland and its surrounding environment in complex scenarios, thus constraining the effectiveness and efficiency of segmentation performance.

B. Vision-Language Models

In recent years, technological advancements have shown the complementary advantages of vision and language in learning and understanding world knowledge. A series of studies also emerged [25], [26], [27], [28], aiming to integrate the intuitive perception of vision with the deep understanding capabilities of language. For instance, Zhang et al. [29] leveraged the powerful reasoning abilities of language to develop a method for small object detection in remote sensing based on text semantic fusion graph reasoning. This method, employing an end-to-end approach, learns and constructs complex and diverse relationships from extensive common-sense knowledge, enabling the detector to cross category boundaries and perform broader and deeper reasoning analysis. Additionally, a few-shot object segmentation method that integrates vision and language was devised [30]. It uses a text knowledge extractor and a cross-modal assembly module to fuse the visual and language modalities, significantly enhancing the accuracy of object segmentation under few-shot conditions. Similarly, in [31], text was taken as a domain-invariant feature that spans across different scenarios, and text descriptions at various granularities were used to guide the model, achieving effective cross-scene hyperspectral image classification. This demonstrates the potential of VLMs in complex scene analysis.

In spite of these efforts, VLMs are seldom used for farmland segmentation. This study centers on farmland segmentation and aims to design a VLM specifically tailored for farmland remote sensing applications. In addition, high-quality image-text pair datasets in VLMs can provide previous knowledge, enabling the model to learn complex mapping relationships between vision and language [32] and achieve efficient downstream tasks.

Presently, high-quality remote sensing image-text pair datasets are insufficient. To this end, Zhang et al. [33] constructed a remote sensing image-text paired dataset by filtering publicly available image-text paired datasets and using a pre-trained VLM to generate captions for those datasets. Additionally, in [34], data scaling was used to convert annotations from remote sensing semantic segmentation datasets and object detection datasets into a unified image annotation data format based on box-to-caption and mask-to-box. This resulted in the generation of a dataset that is 12 times larger than the total size of all available datasets. As the text descriptions directly generated by models may not guarantee the dataset quality, Hu et al. [35] employed a purely manual approach to create a high-quality remote sensing image-text dataset that includes both scene descriptions and object information.

In spite of remarkable progress in image-text dataset construction, datasets that rely on models to automatically generate descriptions still face quality issues, hindering precise and nuanced alignment between images and texts [32]. Although manual annotation offers a way to achieve high data quality, it demands considerable time and effort from domain experts. This results in substantial consumption of valuable resources and inefficiency in the process [36]. Existing remote sensing image-text pair datasets have, furthermore, limited coverage of farmland, leading to a data scarcity dilemma for VLMs when applied to farmland segmentation tasks. It is, therefore, urgent to develop an efficient and high-quality method for constructing FIT datasets, which can minimize the reliance on manual annotation while ensuring the datasets with comprehensive professional knowledge of farmland remote sensing. Such methods will support the research and development of VLMs in farmland segmentation and related fields.

III. Methodology

Existing VLMs lack prior knowledge of farmland remote sensing images. This study constructed a FIT dataset to address this problem. A FSVLM was, furthermore, designed by combining a segmentation model with a multimodal LLM. This section will delve into the underlying principles and processes of constructing the FIT pair dataset and the FSVLM.

A. Construction of FIT Pair Dataset

1) Coverage of the Data Selected for Constructing FIT Pair Dataset: Given the significant impact of varying terrain, climatic conditions, and diverse crop cultivation types on farmland landscapes, this study selects several prefecture-level cities from four provinces in China for in-depth research. They are Shaoyang and Hengyang in Hunan, Meizhou, Maoming, and Zhanjiang in Guangdong, Bozhou in Anhui, and Xishuangbanna in Yunnan. These regions encompass a wide range of climatic zones, from subtropical to temperate, and display diverse topographical features, including plains, plateaus, hills, and mountains, as well as a rich variety of crop species and cultivation patterns. This selection aims to create a comprehensive and complex data research framework, facilitating an in-depth exploration of the characteristics

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:58:22 UTC from IEEE Xplore. Restrictions apply.

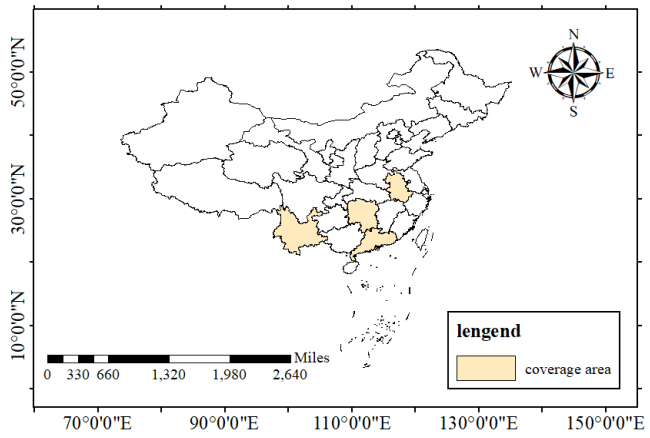


Fig. 1. Coverage of the FIT data used in this study.

of farmland landscapes and examining the segmentation capabilities of VLMs.

The coverage of the FIT dataset is shown in Fig. 1. Hunan, with a farmland area of 36.542 million hectares, accounts for approximately 2.86% of the nation's total farmland. It is characterized by a complex terrain and diverse types of farmlands. Shaoyang and Hengyang in central-south Hunan experience a subtropical monsoon climate, with the primary grain crops being rice, rapeseed, and corn. The farmland area in Guangdong is 19.063 million hectares, accounting for approximately 1.49% of the nation's total farmland. The province is predominantly mountainous and hilly, resulting in significant differentiation in farmland types. Meizhou, situated at the northeastern tip of Guangdong, experiences a subtropical monsoon climate, with rice as the primary grain crop. Maoming and Zhanjiang, located in the southwestern tip of Guangdong, also experience a subtropical monsoon climate. The primary grain crops are rice, tubers, and corn. Anhui province has a cultivated land area of 55.509 million hectares, accounting for approximately 4.35% of the nation's total cultivated land. Its geomorphology is dominated by plains and hills. Bozhou, situated in the northwest of Anhui, experiences a temperate monsoon climate, with the primary grain crops being corn, wheat, and soybeans. Yunnan has a cultivated land area of 52.867 million hectares, accounting for approximately 4.14% of the nation's total cultivated land. Its geomorphology is dominated by mountains and plateaus. Xishuangbanna Dai Autonomous Prefecture, located at the southernmost tip of Yunnan, experiences a tropical monsoon climate, with the primary grain crops being rice and corn.

2) Semi-Automatic Construction of FIT Pairs Dataset: Using the Bigemap software, this study successfully acquired 0.5-m-resolution Google satellite imagery for various months across selected cities. These selected data represent the high diversity of farmland in the imagery and reflect the complex changes in farmland landscapes across different geographical environments. Notably, the series of typical remote sensing images depicts the distribution, morphology, and surrounding environments of farmlands (see Fig. 2). By meticulously describing these images through text, the model can learn the farmland shapes, as well as the closely related facilities, natural vegetation, and man-made structures of farmland in different regions, thereby gaining a more comprehensive understanding of the diversity and complexity of farmland landscapes.

TABLE I
IMAGE DETAILS

Coverage area		Size(pixel)	Number	Collection time
Hunan	Shaoyang and Hengyang	512*512	944	January
Guangdong	Meizhou, Maoming, and Zhanjiang	512*512	5381	February April December
Anhui	Bozhou	512*512	696	November
Yunan	Xishuangbanna	512*512	248	January

The acquired farmland imagery was preprocessed using ENVI software to eliminate distortions caused by atmospheric and other physical factors. The images were then uniformly cropped to a size of 512 × 512 pixels. Next, images without farmland were manually screened and excluded, resulting in a total of 7269 images across all selected regions. Details of those image data are shown in Table I. In order to minimize the time cost of annotation, the segment anything model (SAM) [37] was integrated into Labelme in this study. This allows for semi-automatic mask annotation of the images, enabling us to obtain imagery labels. Some examples of partially annotated labels are illustrated in Fig. 3.

By analyzing farmland images from different regions and considering the regional low segmentation accuracy in existing work, this study summarized the temporal and spatial factors that may affect the accuracy of farmland segmentation, as shown in Table II. These 12 key factors, including temporal and spatial elements, can be classified into intrinsic attributes of farmland and attributes of the surrounding environment. Intrinsic attributes include shooting time, geographical location, landscape characteristics, farmland distribution, farmland shape, internal farmland roads, and terrain. The surrounding environmental factors involve the distribution of water bodies, vegetation, and buildings, as well as the sowing situation of crops.

These 12 factors were input into a farmland description text template, as shown in Fig. 4, to achieve semi-automatic text annotation. The descriptive texts corresponding to these factors were integrated as option controls into Labelme. Then, by combining image visual features, the descriptive text was semi-automatically annotated, with the month and geographic location of the images being automatically obtained from the images. Because of the limitation of image cropping size, some images might omit ground object categories other than farmland. When using a semi-automatic annotation framework to annotate surrounding environmental attributes, a preliminary assessment was, therefore, used to confirm the presence of relevant features, ensuring the accuracy of the descriptive text.

B. VLM for Farmland Segmentation

1) Overall Architecture of FSVLM: This study introduced the FSVLM, which combines a multimodal LLM module with a segmentation model to achieve efficient farmland segmentation. Inspired by Lai et al. [38], the "embedding as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:58:22 UTC from IEEE Xplore. Restrictions apply.

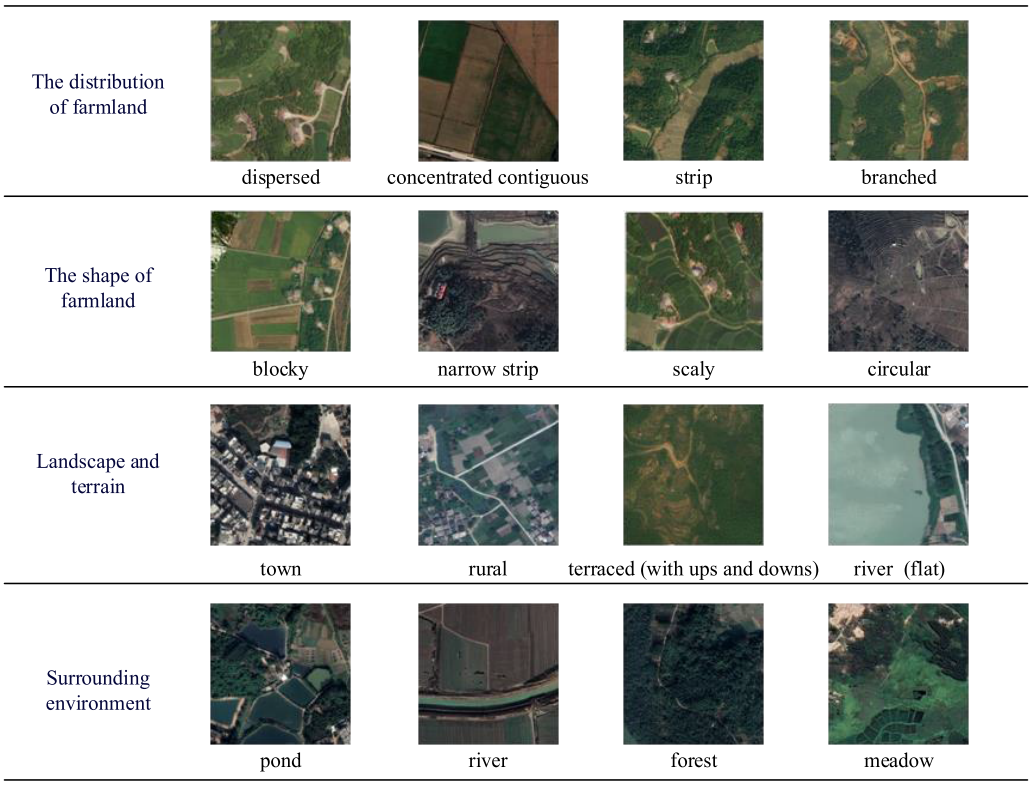


Fig. 2. Typical distribution, shape, and surrounding environment of farmland.

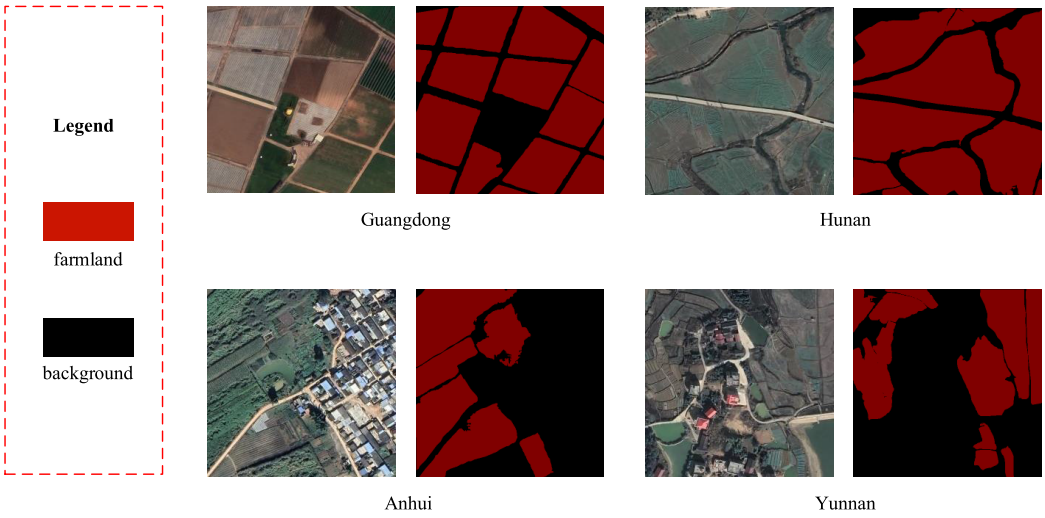


Fig. 3. Examples of partially labeled image data.

mask" paradigm was used in the LLM model. The overall architecture of the model is shown in Fig. 5.

The multimodal LLM branch employs the "embedding-as-mask" paradigm to embed segmentation tokens ⟨SEG⟩ within the semantic tokens of the descriptive text, guiding the model to generate segmentation masks. The input text includes prompt text and descriptive text. The prompt text is a request for outputting a semantic segmentation map, such as "please output the farmland segmentation mask." The descriptive text is included in the constructed FIT. Details are in Section III-A2. Subsequently, the multimodal LLM establishes a relationship between vision and language, directing the segmentation model to generate precise farmland masks. Then, a multilayer perceptron (MLP) is used to perform nonlinear mapping on the output of multimodal predictions to enhance feature expression ability. Finally, the feature sequence obtained by the MLP is sent to the decoder to guide the segmentation model in generating masks.

The segmentation model branch considers that the prompt encoder in the SAM can effectively guide the decoder to generate masks by sparsely encoding text features; thus, in this study, the SAM was used as the segmentation model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:58:22 UTC from IEEE Xplore. Restrictions apply.

TABLE II
FACTORS IMPACTING FARMLAND SEGMENTATION

Factors	Options
Shooting time (month)*	——
Geographic location (latitude and longitude) *	——
Landscape features	rural, town
Distribution of cultivated land	strip, branched, concentrated contiguous, dispersed
Shape of cultivated land	blocky, scaly, narrow strip, broken, circular
Internal roads within cultivated land	crisscrossed in straight lines, curved
Terrain	undulating, flat
Water	a winding small river, large continuous water bodies, scattered blocky ponds, none
Vegetation	a vast forest, scattered forests, scattered trees, meadow, none
Buildings	clustered, scattered, none
Sowing situation	all cultivated land has crops, all cultivated land has no crops, the cultivated land with crops is more than that without crops, the cultivated land with crops is less than that without crops, the cultivated land with crops is approximately equal to the cultivated land without crops

Note: * indicates option automatically obtained by the semi-automatic framework.

This satellite image was taken in [Shooting time], located at [Geographic location] degrees east longitude and [Geographic location] degrees north latitude. It shows a [Landscape features] landscape, with the farmland primarily in a [Distribution of farmland] distribution. The shape of the farmland is characterized by [Shape of the farmland]. The internal roads in the farmland are [Internal roads of farmland], and the terrain is [Terrain]. The water bodies surrounding the farmland mainly consist of [Water]. The vegetation around the farmland mainly consists of [Vegetation]. The buildings around the farmland primarily appear as [Buildings]. The sowing situation indicates that [Sowing situation].

Fig. 4. Text template for farmland description.

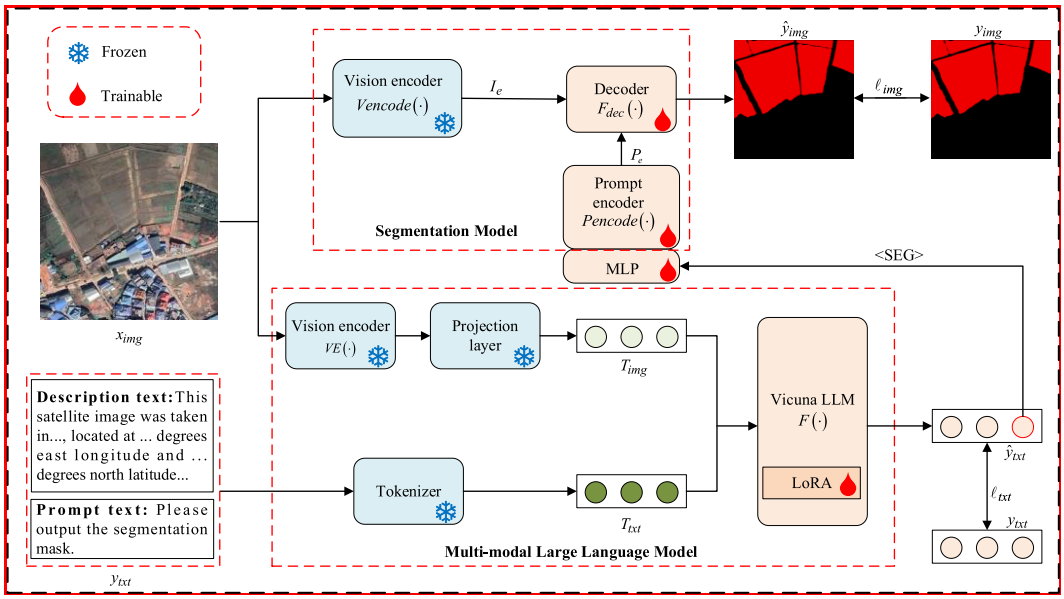


Fig. 5. Overall architecture of FSVLM.

Specifically, in FSVLM, the visual feature encoder within the segmentation model is frozen, whereas the decoder and MLP projection layers are trainable. In order to preserve the generalization ability of the pretrained multimodal LLM, low-rank adaption (LoRA) [39] is, furthermore, employed for efficient fine-tuning.

2) Multimodal LLM of FSVLM: The multimodal LLM module is based on the large language and vision assistant (LLaVA) model [40]. Given the farmland image x_{img} and its query text y_{txt}, the visual features of x_{img} are extracted using a contrastive language-image pretraining (CLIP) vision encoder [41] and a linear projection layer. Those features are converted into a vision token sequence T_{img}, which has the same dimension as the word token sequence T_{txt} in the language model. This process can be expressed as

$$T_{\text{img}} = P\left(\text{VE}\left(x_{\text{img}}, \omega^{\text{mv}}\right)\right) \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:58:22 UTC from IEEE Xplore. Restrictions apply.

where VE(·) is the CLIP vision encoder, ω^{mv} is the corresponding parameter, and $P(\cdot)$ is a linear projection layer.

Subsequently, the vision token sequence T_{img} and the word token sequence T_{txt} corresponding to the input text y_{txt} are processed together by Vicuna [42] to generate a response text $\hat{y}_{txt}$. This process can be expressed as

$$\hat{y}_{txt} = F\left(T_{img}, T_{txt}\right) \tag{2}$$

where $F(\cdot)$ represents the processing procedure of Vicuna. At the same time, a new token ⟨SEG⟩ is added to LLM to represent the request to output semantic segmentation masks. When the query text y_{txt} requires outputting segmentation masks, the generated text contains token ⟨SEG⟩ and places it in the last layer of the embedding layer of LLMs. By processing this embedding layer through an MLP, t_{seg} containing segmentation prompt information can be obtained

$$t_{seg} = \text{MLP}\left(\hat{y}_{txt}, \omega^{t}\right) \tag{3}$$

where MLP(·) represents an MLP, which is a composite operation composed of linear layers, activation functions, and dropout layers. ω^{t} represents the learnable parameters in the MLP.

3) Segmentation Model of FSVLM: SAM was directly used as the segmentation model in this study to simplify the design of FSVLM. SAM consists of three parts: vision encoder, prompt encoder, and decoder. The vision encoder uses the pre-trained vision transformer (ViT) [43] to cope with the input of high-resolution images. During training, FSVLM encodes the input image through a vision encoder to obtain image embedding features. This process can be represented as

$$I_e = V\text{encode}\left(x_{img}, \omega^{v}\right) \tag{4}$$

where x_{img} is the input image, Vencode(·) is the vision encoder, and ω^{v} is the corresponding parameters. I_e is the embedded feature of the output image. In this study, after obtaining the image embedding features, the prompt features input by the multimodal LLM are passed through a prompt encoder. This step yields sparse embedding features. This process can be described as

$$P_e = P\text{encode}\left(t_{seg}, \omega^{P}\right) \tag{5}$$

where t_{seg} is the output of the MLP. Pencode(·) and ω^{P} represent the prompt encoder and corresponding parameters, respectively. Finally, the image embedding features obtained by the vision encoder and the sparse embedding features obtained by the prompt encoder are input into the decoder to obtain the final segmentation mask $\hat{y}_{img}$ as follows:

$$\hat{y}_{img} = F_{dec}(P_e, I_e) \tag{6}$$

where $F_{dec}(\cdot)$ is the decoder, and $\hat{y}_{img}$ represents the final mask generated by the proposed FSVLM.

C. Implementation Details of FSVLM

In the proposed FSVLM, the auto-regressive cross-entropy loss is used to optimize the multimodal LLM, and per-pixel binary cross-entropy loss (BCE) and Dice coefficient loss (DICE) are used to optimize the segmentation model. For each training sample set consisting of the original farmland image x_{img}, the corresponding ground truth y_{img} and query text y_{txt}, the auto-regressive cross-entropy loss [see (7)] is used to measure the loss between the generated response text $\hat{y}_{txt}$ and the query text y_{txt}

$$\ell_{txt} = \text{CE}(\hat{y}_{txt}, y_{txt}) \tag{7}$$

where CE(·) represents the auto-regressive cross-entropy loss function. Additionally, a weighted combination of the BCE and the DICE loss function is used to measure the bias between the generated mask $\hat{y}_{img}$ and the ground truth y_{img}. This process can be formulated as

$$\ell_{mask} = \lambda_1 \text{BCE}\left(\hat{y}_{img}, y_{img}\right) + \lambda_2 \text{DICE}\left(\hat{y}_{img}, y_{img}\right) \tag{8}$$

where BCE(·) and DICE(·), respectively, represent the BCE loss function and the DICE loss function. λ_1 and λ_2 correspond to the initial weights of BCE(·) and DICE(·), respectively. The final overall objective function ℓ can be expressed as

$$\ell = \ell_{txt} + \ell_{mask}. \tag{9}$$

The implementation details of the final model are shown in Algorithm 1.

IV. Experiments and Analysis

A. Experimental Setting

The FSVLM model was trained using an NVIDIA A6000 GPU, and adopted the AdamW optimizer. The learning rate and weight decay were set to $3e^{-4}$ and 0, respectively. Thetraining was conducted over 10 epochs, with each epoch consisting of 1000 steps. The weights λ_1 and λ_2 for the BCE and the DICE were set to 2.0 and 0.5, respectively. The batch size was set to 4. Finally, LLaVA-13B-v-1-1 was used for the multimodal LLM and ViT-H SAM was used for the segmentation model.

B. Dataset and Evaluation Indicators

The datasets used in the experiment include the FIT dataset constructed in this study, as well as LoveDA [44]. The FIT dataset has a spatial resolution of 0.5 m and comprises a total of 7269 farmland images, covering regions in four provinces of China: Hunan, Guangdong, Anhui, and Yunnan. Detailed information about FIT can be found in Section III-A. The LoveDA dataset, constructed by the RSIDEA team, has a spatial resolution of 0.3 m and contains 5987 images covering the cities of Nanjing, Changzhou, and Wuhan in China. The detailed information of the LoveDA dataset is shown in Table III.

This study adopted four metrics commonly used in farmland segmentation research: mean accuracy (mACC), mean intersection over union (mIoU), mean Dice coefficient (mDice), and Recall. mACC refers to the average classification

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:58:22 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Implementation Process of FSVLM

Input:original farmland image x_{img}, ground truth y_{img}, query text y_{txt}.
Output: Prediction Mask $\hat{y}_{img}$.
For epoch $= 0$ **to** 10 **do**
  For step $= 0$ **to**1000 **do**
    1. Using (1), encode the visual features of the input original farmland image x_{img} and perform linear mapping to obtain visual sequence features T_{img} with the same dimension as the word embedding sequence.
    2. Formula (2) is used to process visual sequence feature T_{img} and text embedding token sequence feature T_{txt} to obtain text response features $\hat{y}_{txt}$.
    3. Calculate the loss between the features of real text y_{txt} and predicted text $\hat{y}_{txt}$ using (7), and optimize the model
    4. Extract segmentation prompt feature t_{seg} from text response feature $\hat{y}_{txt}$ using (3).
    5. Encode the original farmland image x_{img} by (4) and convert it into image embedding features I_e.
    6. Encode the semantic segmentation prompt feature t_{seg} obtained in step **4** as a sparse embedding feature P_e using (5).
    7. Decode I_e obtained in step **5** and P_e obtained in step **6** by (6) to obtain the prediction mask $\hat{y}_{img}$.
    8. Calculate the pixel-by-pixel binary cross entropy loss and DICE loss between ground truth y_{img} and prediction mask $\hat{y}_{img}$ according to (8), and perform back-propagation to optimize the model.
  End for
End for

TABLE III
DETAILED INFORMATION OF THE LOVEDA DATASET

Key element	Detail
Coverage	Nanjing, Changzhou, Wuhan
Scene	Rural, Urban
Spatial resolution	0.3m
Size	1024*1024 pixel
Classes	7
Number	5987

accuracy of pixels across all classes. mIoU measures the average ratio of model intersection to union, serving as a widely accepted metric in semantic segmentation tasks. mDice evaluates set similarity. Recall assesses the proportion of positive samples that are correctly identified and segmented by the model.

C. Comparative Experiments and Analysis

In order to validate the effectiveness of the FSVLM model, we compared it with six label-learning based semantic segmentation methods: Deeplabv3, SegFormer [45], MaskFormer [46], Mask2Former [47], SegNeXt [48], and DDRNet [49].

Deeplabv3 employs an encoder-decoder architecture that combines the advantages of spatial pyramid pooling module and encoder-decoder structure, improving the perception ability and segmentation accuracy of multiscale targets. SegFormer achieves a balance between high accuracy and low computational cost through efficient feature extraction and fusion strategies coupled with a simple decoder design. MaskFormer is a unified framework designed to address image segmentation tasks, including semantic segmentation and instance segmentation. Mask2Former employs a transformer architecture to capture global relationships within images, enhancing the segmentation accuracy and efficiency of the model through a series of optimization strategies. SegNeXt proposes a novel multiscale convolutional attention (MSCA) module, which improves upon traditional convolutional structures and achieves performance that surpasses the transformer to a certain extent. DDRNet combines a network architecture with branches of different resolutions and a deep aggregation pyramid pooling module, significantly boosting inference speed while maintaining high accuracy.

All methods were trained and tested using the FIT dataset constructed in this study, with a training-to-testing sample ratio of 7:3. Notably, only the FSVLM is a VLM model. The other methods only use the original farmland images and their corresponding ground truth from the FIT for training and evaluation.

The results are shown in Tables IV–VII. Overall, the FSVLM demonstrates better performance over other methods across several evaluation metrics, highlighting the strong potential of FSVLM tasks. Specifically, the FSVLM demonstrates significant advantages in mACC, indicating that the model achieves greater precision in segmenting farmland areas and reduces the possibility of misjudgment between farmland and nonfarmland regions. Additionally, FSVLM also demonstrated significant improvements in mIoU and mDice. As an important metric for measuring the overlap between the predicted and real areas, a high mIoU value indicates the high accuracy of FSVLM in delineating farmland boundaries. The mDice coefficient further strengthens this finding, demonstrating the stability and robustness of FSVLM in complex scenarios; however, as shown in Tables VI and VII, the recall of FSVLM in the Hunan region is slightly lower than that of Mask2Former. The recall rate reflects the model's ability to identify all positive samples. The slight deficiency of FSVLM may stem from its preference for high-confidence predictions, resulting in the failure to fully capture all farmland information in some boundaries or ambiguous areas. In spite of this, the FSVLM remains superior overall in farmland segmentation tasks, especially in scenarios that require high accuracy and stability.

In addition, to assess the performance of FSVLM in farmland segmentation, the segmentation results of all methods across different provinces were visualized, as shown in Figs. 6–9. The FSVLM segmentation results show a high degree of alignment with the ground truth, exhibiting extraordinary similarity in both overall layout and detailed texture. This demonstrates the ability of FSVLM in capturing farmland features and distinguishing nonfarmland areas. Despite these impressive results, minor instances of "salt and pepper noise" are evident in some segmentation maps. This may be due to the complexity of the image data or the slight

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:58:22 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
FARMLAND SEGMENTATION ACCURACY OF DIFFERENT METHODS IN ANHUI

Evaluating indicator (%)	Method						
	Deeplabv3	SegFormer	MaskFormer	Mask2Former	SegNeXt	DDRNet	FSVLM
mACC	87.86	87.73	89.83	87.24	87.41	86.97	**92.58**
mIoU	73.75	80.31	74.64	74.02	73.26	73.59	**80.69**
mDice	86.28	88.92	80.05	84.93	88.41	87.19	**92.53**
Recall	87.54	88.50	89.57	89.24	87.46	86.57	**90.08**

TABLE V
FARMLAND SEGMENTATION ACCURACY OF DIFFERENT METHODS IN GUANGDONG

Evaluating indicator (%)	Method						
	Deeplabv3	SegFormer	MaskFormer	Mask2Former	SegNeXt	DDRNet	FSVLM
mACC	71.09	84.19	71.53	84.01	82.35	70.67	**89.98**
mIoU	65.91	73.28	66.42	72.39	74.11	56.39	**79.05**
mDice	75.50	83.44	77.24	82.78	83.98	74.11	**88.56**
Recall	72.54	86.64	72.89	86.98	83.65	69.64	**89.74**

TABLE VI
FARMLAND SEGMENTATION ACCURACY OF DIFFERENT METHODS IN HUNAN

Evaluating indicator (%)	Method						
	Deeplabv3	SegFormer	MaskFormer	Mask2Former	SegNeXt	DDRNet	FSVLM
mACC	86.95	86.87	85.14	87.92	85.42	82.09	**88.49**
mIoU	76.34	77.22	74.52	78.26	74.71	71.59	**78.57**
mDice	85.87	86.78	84.92	88.76	85.07	81.38	**89.26**
Recall	86.45	86.46	85.14	**88.96**	84.74	80.62	87.58

TABLE VII
FARMLAND SEGMENTATION ACCURACY OF DIFFERENT METHODS IN YUNNAN

Evaluating indicator (%)	Method						
	Deeplabv3	SegFormer	MaskFormer	Mask2Former	SegNeXt	DDRNet	FSVLM
mACC	80.92	84.55	82.98	85.98	77.79	77.32	**89.35**
mIoU	68.54	71.73	70.93	75.49	66.15	65.51	**78.61**
mDice	80.09	83.41	82.83	85.89	79.14	78.08	**90.01**
Recall	81.20	83.92	82.45	**87.96**	78.59	79.37	87.16

model deviation in dealing with specific boundary cases. These imperfections suggest areas where the model could be further improved.

In summary, the FSVLM algorithm has demonstrated comprehensive and significant advantages over other methods in farmland segmentation based on the FIT dataset. It enhances both the accuracy and reliability of farmland segmentation and provides strong technical support and reference for subsequent research and applications.

D. Generalization Performance of FSVLM

The irregularity and variability in the spatial structure and distribution of farmland pose significant challenges to the model's generalization. In order to evaluate the adaptability of the FSVLM model to untrained data, this study compared the farmland segmentation performance of all methods trained on the FIT dataset when applied to new regions. Specifically, the LoveDA dataset was used as an independent test benchmark, encompassing farmland samples from regions with distinct regional characteristics, including Nanjing, Wuhan, and Changzhou. These areas were excluded during the model training phase, ensuring the unknown and challenging testing environment. Given that this study mainly focused on farmland segmentation tasks, our evaluation was confined to the rural areas within the LoveDA dataset. The dataset includes a total of 2358 images of the rural areas. The experimental results are shown in Table VIII.

As the results show, the FSVLM model exhibits excellent farmland segmentation performance, even in completely unfamiliar geographical areas, outperforming other methods. These findings not only demonstrate the model's effectiveness in segmenting farmland across geographical differences, but also show its potential for application in farmland recognition in unknown environments. This further proves the strong generalization and transfer capabilities of FSVLM in farmland segmentation.

E. Ablation Experiments

Presently, little work has been done on evaluating the effectiveness of different types of descriptive texts in improving the performance of VLMs in specific tasks. In order to fill this gap, this study designed a set of ablation experiments to explore the impact of different descriptive texts on the farmland segmentation performance of FSVLM.

In the experiments, each sample's language modality comprises both prompt text and descriptive text. From the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:58:22 UTC from IEEE Xplore. Restrictions apply.

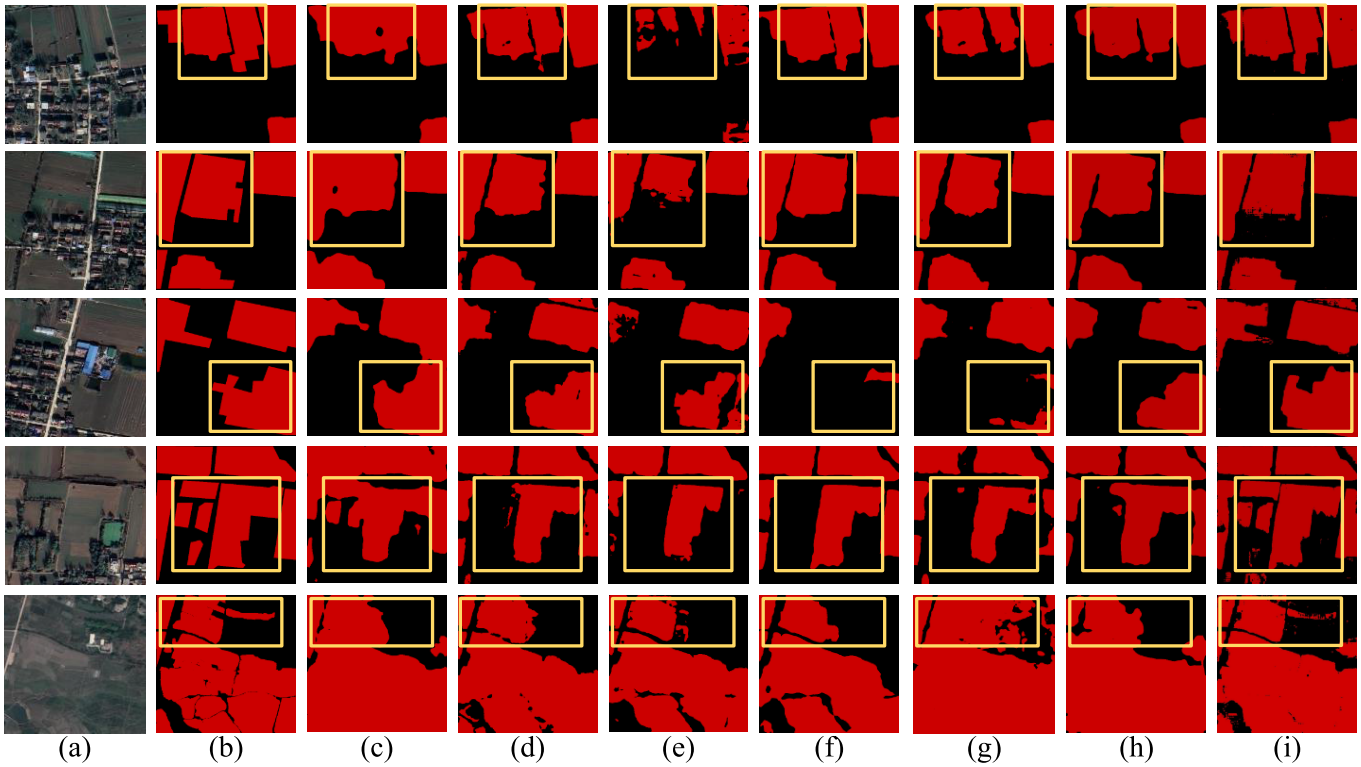


Fig. 6. Farmland segmentation results of different methods in Anhui. (a) Original image, (b) ground truth, (c) Deeplabv3, (d) SegFormer, (e) MaskFormer, (f) Mask2Former, (g) SegNeXt, (h) DDRNet, and (i) FSVLM.

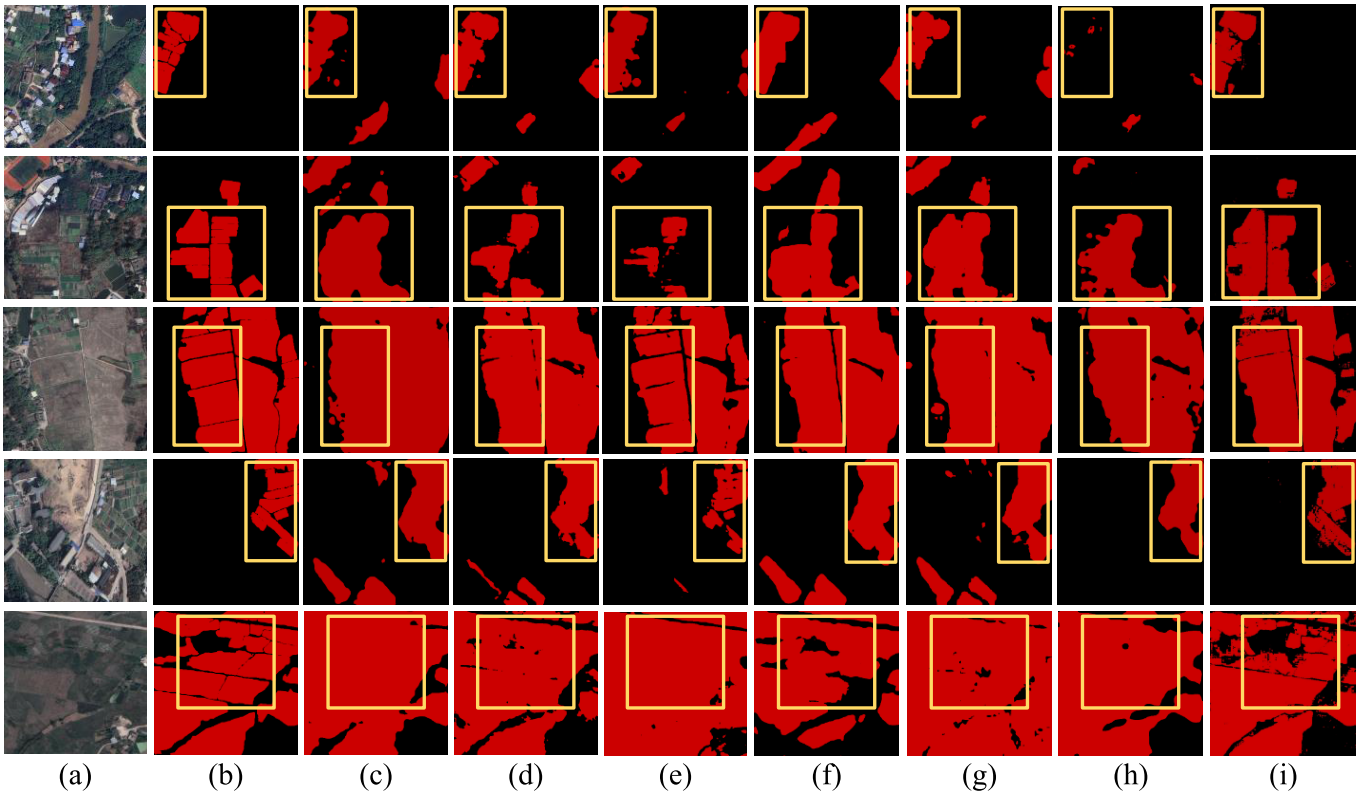


Fig. 7. Farmland segmentation results of different methods in Guangdong. (a) Original image, (b) ground truth, (c) Deeplabv3, (d) SegFormer, (e) MaskFormer, (f) Mask2Former, (g) SegNeXt, (h) DDRNet, and (i) FSVLM.

perspective of textual semantic connotation, descriptive texts are defined into two categories: those detailing the farmland attributes and those describing the surrounding environments. The impact of adding different semantic descriptive texts

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:58:22 UTC from IEEE Xplore. Restrictions apply.

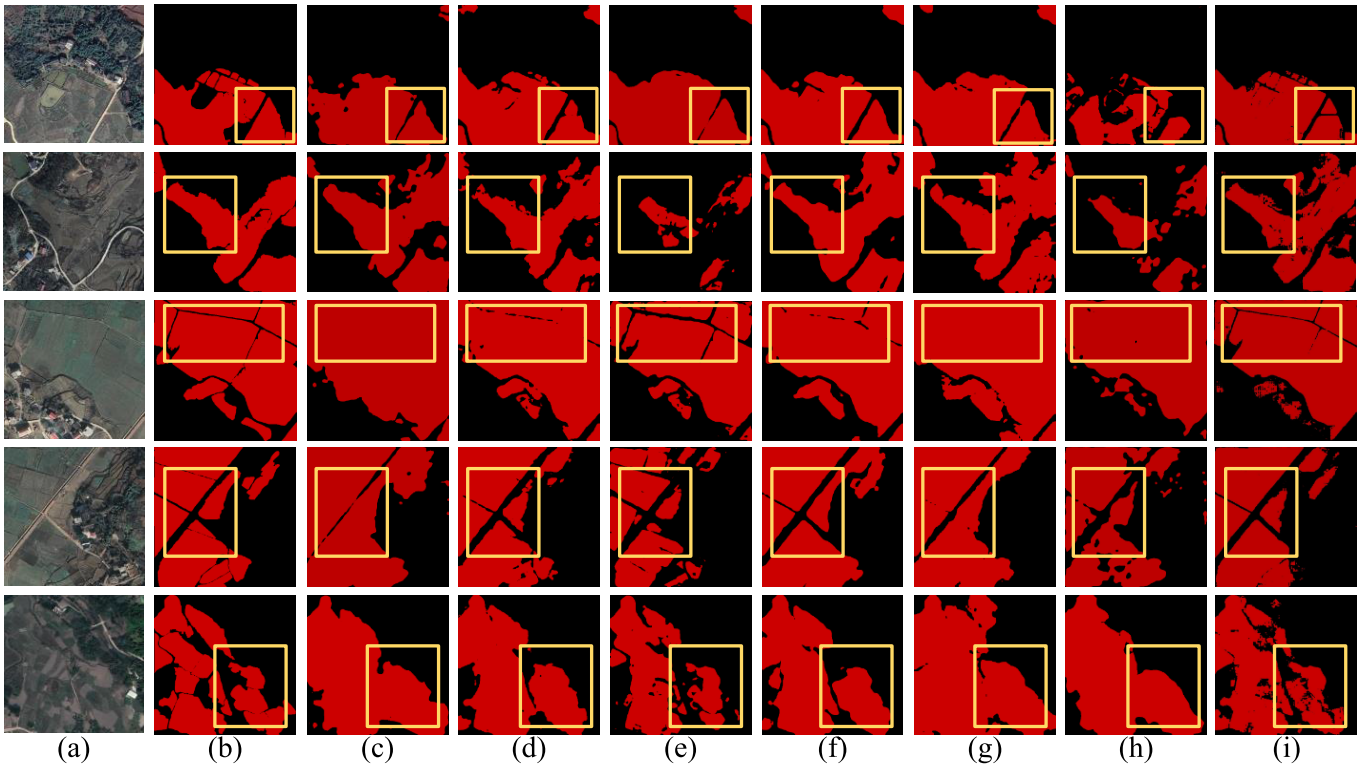


Fig. 8. Farmland segmentation results of different methods in Hunan. (a) Original image, (b) ground truth, (c) Deeplabv3, (d) SegFormer, (e) MaskFormer, (f) Mask2Former, (g) SegNeXt, (h) DDRNet, and (i) FSVLM.

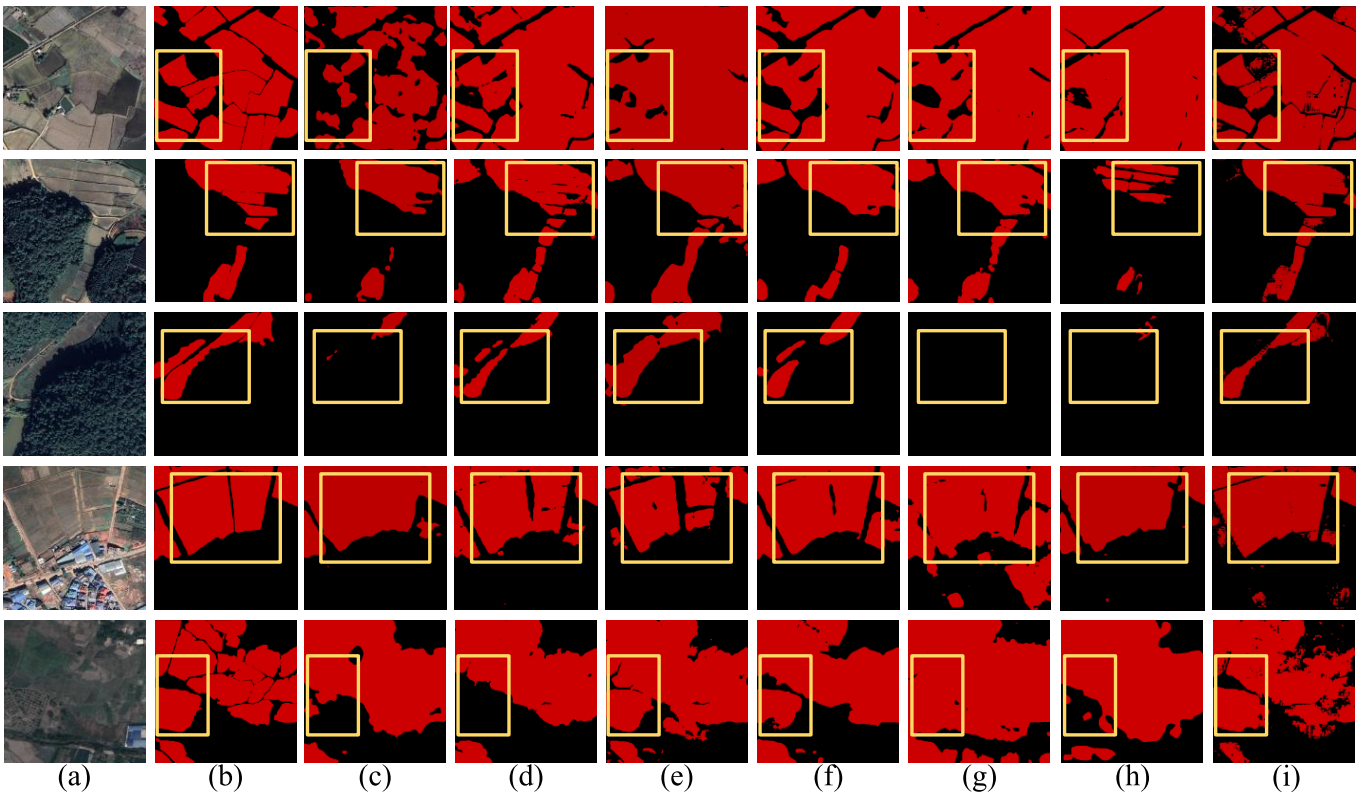


Fig. 9. Farmland segmentation results of different methods in Yunnan. (a) Original image, (b) ground truth, (c) Deeplabv3, (d) SegFormer, (e) MaskFormer, (f) Mask2Former, (g) SegNeXt, (h) DDRNet, and (i) FSVLM.

on the farmland segmentation performance of FSVLM was compared, and the results are shown in Table IX.

Based on the analysis of the results in Table IX, the following conclusion can be drawn.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:58:22 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
FARMLAND SEGMENTATION RESULTS OF DIFFERENT METHODS ON THE LOVEDA

Evaluating indicator (%)	Method						
	Deeplabv3	SegFormer	MaskFormer	Mask2Former	SegNeXt	DDRNet	FSVLM
mACC	50.01	80.42	76.59	82.47	49.99	61.89	**86.78**
mIoU	15.00	70.61	65.99	67.03	35.00	46.32	**70.56**
mDice	23.09	81.48	78.64	79.92	41.18	61.73	**84.26**
Recall	50.01	78.67	77.93	81.75	50.09	62.53	**86.05**

TABLE IX
RESULTS OF ABLATION EXPERIMENTS

Evaluating indicator (%)	A	B	C	D
mACC	87.44	89.05	88.62	**90.10**
mIoU	75.72	78.48	76.35	**79.03**
mDice	84.47	88.58	86.31	**90.09**
Recall	86.75	88.26	87.21	**89.89**

Note:
A: Add prompt text, but not descriptive text
B: Add prompt text and partial descriptive text (remove descriptions related to the surrounding environment of farmland)
C: Add prompt text and partial descriptive text (excluding descriptions of the shape, distribution, and other attributes of the cultivated land)
D: Add prompt text and complete descriptive text

1) *Effects of descriptive texts:* Introducing descriptive texts can significantly improve the accuracy of farmland segmentation in FSVLM, as evidenced by comparing Experiment A with Experiments B–D. This result proves the value of the semantic information embedded in the descriptive text, providing a new dimension to deeply understand and extract the complex features of farmland.
2) *Effects of describing key attributes:* A comparison of Experiments B and C with Experiment D reveals a notable decline in model accuracy when removing descriptions of core attributes such as farmland shape and distribution. This result highlights the role of key descriptions of farmland's attributes in improving the FSVLM's understanding ability, further verifying the importance of these attributes for accurately extracting farmland features.
3) *Effects of semantic contexts:* Comparing Experiment D with Experiments A–C, we found that the FSVLM's accuracy exhibits a significant improvement when it receives complete descriptive texts. This phenomenon reveals the intimate contextual semantic connections among farmland descriptive texts. These connections generate a remarkable synergistic effect in optimizing the FSVLM's performance, collectively facilitating a comprehensive and accurate understanding of farmland features by the FSVLM.

In summary, this ablation experiment explored the effectiveness of different descriptive texts in improving the performance of FSVLM in farmland segmentation. The analysis, which considers the accuracy improvement got by descriptive text, the necessity of describing key attributes, and the effect of semantic context, elucidates the crucial role of specific types of language in the field segmentation task of VLMs. These findings provide valuable practical insights for the optimization and application of VLMs in farmland segmentation tasks in the future.

V. CONCLUSION

This study focused on farmland segmentation across different regions of China. First, a semi-automatic text description framework was proposed by obtaining high-resolution remote sensing images and combining 12 key factors that affect farmland segmentation accuracy. Then, the automated text description framework was used to construct a FIT dataset. Finally, vision and language modalities were fused by the proposed FSVLM for remote sensing farmland information extraction. The performance of FSVLM was better compared to other farmland segmentation methods. Specifically, the FSVLM model can more effectively distinguish between farmland and nonfarmland areas and demonstrates stronger adaptability and better extraction performance in complex terrain environments. This study demonstrated that the farmland segmentation method driven by image language models can significantly improve farmland segmentation accuracy. This study also explored the importance and synergistic effect of different language descriptions on improving model performance, providing a reference for applying VLM to large-scale farmland segmentation.

Although FSVLM demonstrates excellent segmentation capabilities in most scenarios, issues such as missegmentation still arise in extremely complex farmland environments. This may be due to the scattered distribution of various ground objects in these complex agricultural scenes, as well as the model's limitations in handling text length, which leads to insufficient spatial alignment grained between the images and text in the FSVLM.

Future work will, therefore, focus on optimizing the image-text spatial alignment, further enhancing the model's ability to understand and process complex farmland environments. This will help reduce segmentation errors and lay a solid foundation for achieving precise and efficient farmland segmentation across a broader range of application scenarios.

REFERENCES

[1] M. Weiss, F. Jacob, and G. Duveiller, "Remote sensing for agricultural applications: A meta-review," *Remote Sens. Environ.*, vol. 236, Jan. 2020, Art. no. 111402.
[2] C. Shi, X. Zhang, L. Wang, and Z. Jin, "A lightweight convolution neural network based on joint features for remote sensing scene image classification," *Int. J. Remote Sens.*, vol. 44, no. 21, pp. 6615–6641, Nov. 2023.
[3] C. Shi, J. Chen, and L. Wang, "Hyperspectral image classification based on a novel lush multi-layer feature fusion bias network," *Expert Syst. Appl.*, vol. 247, Aug. 2024, Art. no. 123155.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:58:22 UTC from IEEE Xplore. Restrictions apply.

[4] H. Wu, C. Shi, L. Wang, and Z. Jin, "A cross-channel dense connection and multi-scale dual aggregated attention network for hyperspectral image classification," *Remote Sens.*, vol. 15, no. 9, p. 2367, Apr. 2023.

[5] A. Shrestha and A. Mahmood, "Review of deep learning algorithms and architectures," *IEEE Access*, vol. 7, pp. 53040–53065, 2019.

[6] W. Liu et al., "Farmland parcel mapping in mountain areas using time-series SAR data and VHR optical images," *Remote Sens.*, vol. 12, no. 22, p. 3733, Nov. 2020.

[7] D. Zhang et al., "A generalized approach based on convolutional neural networks for large area cropland mapping at very high resolution," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111912.

[8] A. Taravat, M. P. Wagner, R. Bonifacio, and D. Petit, "Advanced fully convolutional networks for agricultural field boundary detection," *Remote Sens.*, vol. 13, no. 4, p. 722, Feb. 2021.

[9] S. Wang, G. Sun, B. Zheng, and Y. Du, "A crop image segmentation and extraction algorithm based on mask RCNN," *Entropy*, vol. 23, no. 9, p. 1160, Sep. 2021.

[10] R. Lu, N. Wang, Y. Zhang, Y. Lin, W. Wu, and Z. Shi, "Extraction of agricultural fields via DASFNet with dual attention mechanism and multi-scale feature fusion in south Xinjiang, China," *Remote Sens.*, vol. 14, no. 9, p. 2253, May 2022.

[11] Y. Wang, L. Gu, T. Jiang, and F. Gao, "MDE-UNet: A multitask deformable UNet combined enhancement network for farmland boundary segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[12] M. Zhu, M. Yao, Y. He, Y. He, and B. Wu, "Studies on high-resolution remote sensing sugarcane field extraction based on deep learning," *IOP Conf. Ser., Earth Environ. Sci.*, vol. 237, Mar. 2019, Art. no. 032046.

[13] X.-F. Kuang, J. Guo, H.-Y. Wang, and H. Wang, "Agricultural field boundary delineation using a cascaded deep network model from polarized SAR and multispectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 7228–7247, 2023.

[14] Y. Xie, S. Zheng, H. Wang, Y. Qiu, X. Lin, and Q. Shi, "Edge detection with direction guided postprocessing for farmland parcel extraction," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3760–3770, 2023.

[15] Y. Xu, Z. Zhu, M. Guo, and Y. Huang, "Multiscale edge-guided network for accurate cultivated land parcel boundary extraction from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4501020, doi: 10.1109/TGRS.2023.3344122.

[16] X. Li, C. Wen, Y. Hu, Z. Yuan, and X. X. Zhu, "Vision-language models in remote sensing: Current progress and future trends," *IEEE Geosci. Remote Sens. Mag.*, vol. 12, no. 2, pp. 32–66, Jun. 2024.

[17] I. Lauriola, A. Lavelli, and F. Aiolli, "An introduction to deep learning in natural language processing: Models, techniques, and tools," *Neurocomputing*, vol. 470, pp. 443–456, Jan. 2022.

[18] S. Wang, N. Ding, N. Lin, J. Zhang, and C. Zong, "Language cognition and language computation—Human and machine language understanding," *SCIENTIA SINICA Informationis*, vol. 52, no. 10, pp. 1748–1774, 2022.

[19] W. X. Zhao et al., "A survey of large language models," 2023, *arXiv:2303.18223*.

[20] J. Zhang, J. Huang, S. Jin, and S. Lu, "Vision-language models for vision tasks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5625–5644, Aug. 2024.

[21] C. Persello, V. A. Tolpekin, J. R. Bergado, and R. A. de By, "Delineation of agricultural fields in smallholder farms from satellite images using fully convolutional networks and combinatorial grouping," *Remote Sens. Environ.*, vol. 231, Sep. 2019, Art. no. 111253.

[22] C. Shi, H. Wu, and L. Wang, "CEGAT: A CNN and enhanced-GAT based on key sample selection strategy for hyperspectral image classification," *Neural Netw.*, vol. 168, pp. 105–122, Nov. 2023.

[23] R. Cheng, Y. Wei, and M. Lu, "Cropland field extraction based on ensemble deep learning model," *Chin. J. Agricult. Resour. Regional Planning*, vol. 43, no. 7, pp. 273–281, 2022.

[24] Q. Li, D. Zhang, Y. Pan, and J. Dai, "High-resolution cropland extraction in Shandong province using MPSPNet and UNet network," *Nat. Remote Sens. Bull.*, vol. 27, no. 2, pp. 471–491, 2023.

[25] X. Sun et al., "RingMo: A remote sensing foundation model with masked image modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5612822.

[26] T. Wei, W. Yuan, J. Luo, W. Zhang, and L. Lu, "VLCA: Vision-language aligning model with cross-modal attention for bilingual remote sensing image captioning," *J. Syst. Eng. Electron.*, vol. 34, no. 1, pp. 9–18, Feb. 2023.

[27] M. M. A. Rahhal, M. A. Bencherif, Y. Bazi, A. Alharbi, and M. L. Mekhalfi, "Contrasting dual transformer architectures for multi-modal remote sensing image retrieval," *Appl. Sci.*, vol. 13, no. 1, p. 282, Dec. 2022.

[28] Y. Zhan, Z. Xiong, and Y. Yuan, "RSVG: Exploring data and models for visual grounding on remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5604513, doi: 10.1109/TGRS.2023.3250471.

[29] S. Zhang et al., "Text semantic fusion relation graph reasoning for few-shot object detection on remote sensing images," *Remote Sens.*, vol. 15, no. 5, p. 1187, Feb. 2023.

[30] X. Lu et al., "Few-shot object detection in aerial imagery guided by text-modal knowledge," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5604719.

[31] Y. Zhang, M. Zhang, W. Li, S. Wang, and R. Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501312, doi: 10.1109/TGRS.2022.3233885.

[32] L. Chen et al., "ShareGPT4 V: Improving large multi-modal models with better captions," 2023, *arXiv:2311.12793*.

[33] Z. Zhang, T. Zhao, Y. Guo, and J. Yin, "RS5M and GeoRSCLIP: A large scale vision-language dataset and a large vision-language model for remote sensing," 2023, *arXiv:2306.11300*.

[34] F. Liu et al., "RemoteCLIP: A vision language foundation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5622216, doi: 10.1109/TGRS.2024.3390838.

[35] Y. Hu, J. Yuan, C. Wen, X. Lu, and X. Li, "RSGPT: A remote sensing vision language model and benchmark," 2023, *arXiv:2307.15266*.

[36] M. Zhou et al., "Comparison of data annotation approaches using dependency tree annotation as a case study," *J. Tsinghua Univ. (Sci. Technol.)*, vol. 62, no. 5, pp. 908–916, 2022.

[37] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[38] X. Lai et al., "LISA: Reasoning segmentation via large language model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 9579–9589.

[39] X. Lai et al., "LISA: Reasoning segmentation via large language model," 2023, *arXiv:2308.00692*.

[40] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," 2023, *arXiv:2304.08485*.

[41] A. Radford et al., "Learning transferable vision models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.

[42] W.-L. Chiang et al. (2023). Vicuna: An Open-Source Chatbot Impressing GPT-4 with 90%* ChatGPT Quality. LMSYS Org. [Online]. Available: https://lmsys.org/blog/2023-03-30-vicuna/

[43] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.

[44] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.

[45] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.

[46] B. Cheng, A. G. Schwing, and A. Kirillov, "Per-pixel classification is not all you need for semantic segmentation," 2021, *arXiv:2107.06278*.

[47] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1290–1299.

[48] M.-H. Guo, C.-Z. Lu, Q. Hou, Z. Liu, M.-M. Cheng, and S.-M. Hu, "SegNeXt: Rethinking convolutional attention design for semantic segmentation," 2022, *arXiv:2209.08575*.

[49] H. Pan, Y. Hong, W. Sun, and Y. Jia, "Deep dual-resolution networks for real-time and accurate semantic segmentation of traffic scenes," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 3, pp. 3448–3460, Mar. 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:58:22 UTC from IEEE Xplore. Restrictions apply.